



**SCHOOL OF
ENGINEERING**

DAYANANDA SAGAR UNIVERSITY

Bachelor of Technology in
COMPUTER SCIENCE AND ENGINEERING
(ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

**FSD PROJECT
JOB PORTAL SYSTEM**

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CERTIFICATE

This is to certify that the fsd project (24AM2406) work titled “JOB PORTAL SYSTEM ” is carried out by Puneeth Reddy bearing USN :ENG24AM0248, M.Arjun Reddy bearing USN :ENG24AM0348 and Raviteja.L bearing USN : ENG24AM0075 and Girish Babu.MB bearingUSN : ENG24AM0176 Bonafede student of Bachelor of Technology in Computer Science and Engineering (AI&ML) at the School of Engineering, Dayananda Sagar University, Bangalore in partial fulfillment for the award of degree in Bachelor of Technology in Computer Science and Engineering, during the year 2025-2026.

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DECLARATION

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LIST OF ABBREVIATIONS

HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
JS	JavaScript
API	Application Programming Interface
DBMS	Database Management System
SDLC	Software Development Life Cycle

LIST OF FIGURES

FIG 1	system architecture diagram
FIG 2	view of website
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ABSTRACT

The Job Portal System is a full-stack web application designed to simplify and automate the recruitment process for employers and job seekers. The system provides a centralized digital platform where users can register, log in, search for job opportunities, and apply for jobs by uploading their resumes. Employers can post job vacancies, review applications, and manage recruitment processes efficiently.

Administrators can monitor system activities and maintain data integrity. Traditional recruitment methods involve manual submission of resumes, physical visits to organizations, and lengthy communication processes between candidates and employers. These methods are time-consuming, inefficient, and prone to human errors. With the increasing demand for digital solutions, organizations require automated systems that can manage recruitment processes quickly and accurately.

The Job Portal System is developed using modern full-stack technologies, including HTML and CSS for the frontend interface, Java Spring Boot for backend development, and MySQL for database management. The system supports essential features such as secure user registration and login authentication, job posting, job searching, application submission, resume upload and download, and application tracking.

The implementation of this system improves recruitment efficiency, reduces administrative workload, and ensures secure data management. The project demonstrates the practical application of full-stack development concepts and provides a scalable solution for managing recruitment processes in various organizations.

CHAPTER 1 INTRODUCTION

In today's digital era, technology has transformed the way organizations manage their operations, including recruitment and hiring processes. Online job portals have become essential tools for connecting job seekers and employers efficiently. These systems provide easy access to job opportunities and simplify the process of submitting job applications.

A Job Portal System is a web-based application that allows employers to advertise job vacancies and job seekers to apply for jobs through an online platform. The system eliminates manual paperwork and reduces the time required to manage recruitment activities. It also improves communication between employers and candidates by providing real-time access to job information.

The primary objective of this project is to develop a Job Portal System using modern web technologies that provide a user-friendly interface and secure backend processing. The system enables users to manage job postings, applications, and resumes efficiently. It also demonstrates the integration of frontend, backend, and database technologies in a real-world application.

CHAPTER 2 PROBLEM DEFINITION

Traditional job recruitment systems rely heavily on manual processes such as submitting resumes physically or sending emails to employers. These methods are slow, inefficient, and difficult to manage when the number of applicants increases. Employers often face challenges in organizing candidate information, tracking applications, and maintaining recruitment records.

Job seekers also experience difficulties in finding suitable job opportunities and submitting applications quickly. They may need to visit multiple organizations or

search through various sources to find job openings. This process consumes time and effort and may lead to missed opportunities.

Therefore, there is a need for an automated system that simplifies recruitment processes and improves efficiency. The proposed Job Portal System addresses these challenges by providing an online platform that allows users to manage job postings and applications easily. The system ensures secure data storage, quick access to information, and efficient communication between employers and job seekers.

CHAPTER 3 LITERATURE SURVEY

Several job portal systems have been developed to manage recruitment processes and provide job opportunities to candidates. Popular job portals such as Naukri, Indeed, and LinkedIn offer features such as job search, resume upload, and candidate management. These platforms use web technologies to store user data, manage applications, and provide job recommendations.

Research studies indicate that web-based job portals significantly improve recruitment efficiency and reduce hiring time. Many existing systems use technologies such as PHP, Java, and MySQL for backend development and HTML, CSS, and JavaScript for frontend design. However, some systems have limitations such as complex interfaces, high subscription costs, and limited customization options.

The proposed Job Portal System uses modern technologies such as Spring Boot and MySQL to provide a scalable, secure, and user-friendly recruitment platform. The system also supports resume upload and download functionality, making it more practical for real-world applications.

CHAPTER 4 PROJECT DESCRIPTION

. The Job Portal System is a web-based application designed to manage job postings and job applications efficiently. The system provides different functionalities for job seekers, employers, and administrators.

The system allows users to create accounts and log in securely using authentication mechanisms. Employers can post job vacancies by entering job details such as job title, company name, and job description. Job seekers can view available job listings and apply for jobs by submitting their resumes.

The system also provides application management features that allow administrators to view and manage job applications. All user and job information is stored in a centralized database, ensuring easy access and efficient data management.

CHAPTER 5 PROPOSED DESIGN

The Job Portal System follows a three-tier architecture that separates the system into three layers:

Presentation Layer

Application Layer

Database Layer

The Presentation Layer is responsible for displaying the user interface and collecting user input. This layer is developed using HTML and CSS. The Application Layer processes user requests and performs business logic operations. This layer is implemented using Java Spring Boot.

The Database Layer stores user data, job information, and application records. This layer uses MySQL for data storage and retrieval.

System Flow:

User



Frontend(HTML,CSS)



Controller(SpringBoot)



Repository(JPA)



Database (MySQL)

This architecture ensures scalability, security, and efficient data processing.

CHAPTER 6 SYSTEM ARCHITUCTURE

6 Model Architecture (Detailed)

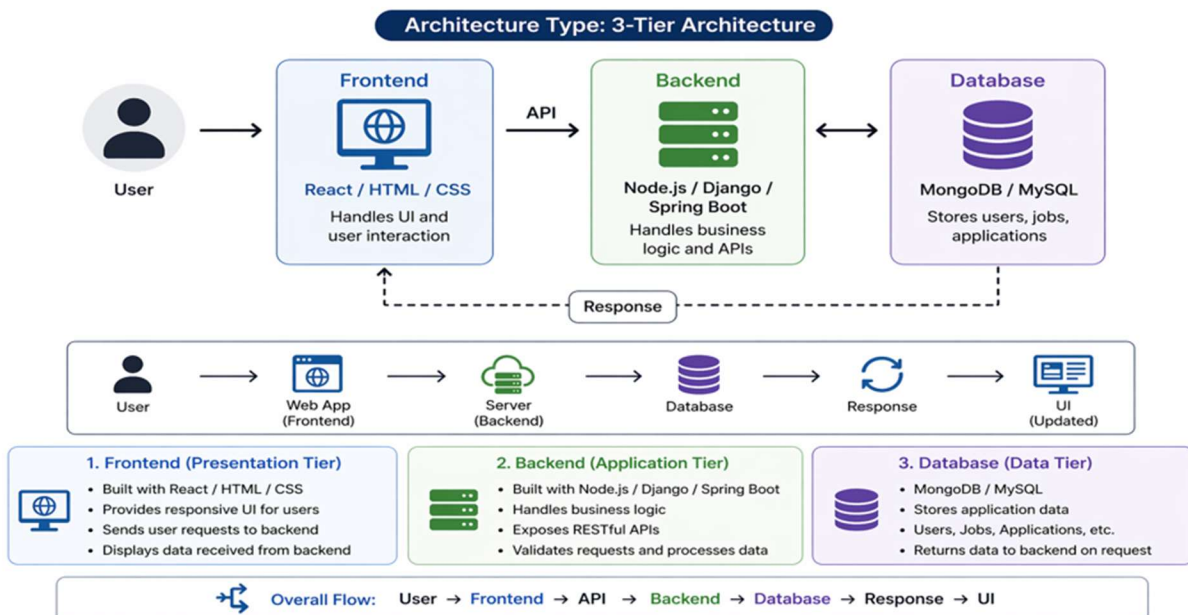


Fig 1:system architecture diagram

System Architecture Explanation (3-Tier Architecture)

The system uses a **3-Tier Architecture**, which divides the application into **three separate layers**. Each layer has a specific responsibility, making the system **scalable, secure, and easy to maintain**.

The Three Tiers are:

1. **Frontend (Presentation Tier)**
2. **Backend (Application Tier)**
3. **Database (Data Tier)**

Frontend (Presentation Tier)

What it is

The **Frontend** is the part of the system that the **user directly interacts with**.

Technologies Used

- React
- HTML
- CSS
- JavaScript

Responsibilities

- Displays the user interface (UI)
- Accepts user input
- Sends requests to the backend
- Shows data received from the server

Example

When a user:

- Opens the website
- Fills a job application form
- Clicks "Submit"

The **Frontend** sends that data to the **Backend**.

Backend (Application Tier)

What it is

The **Backend** is the **brain of the system**.

It processes requests and applies business logic.

Technologies Used

- Node.js
- Django
- Spring Boot

Responsibilities

- Handles business logic

- Processes user requests
- Validates data
- Communicates with the database
- Sends responses back to the frontend

Example

When the user submits a form:

Backend will:

1. Receive the request
2. Check if data is valid
3. Store the data in the database
4. Send confirmation to the frontend

Database (Data Tier)

What it is

The **Database** stores all the system data permanently.

Technologies Used

- MongoDB
- MySQL

Responsibilities

- Stores data
- Retrieves data
- Updates records
- Deletes records

Example Data Stored

- Users
- Jobs
- Applications
- Login details

CHAPTER 7 METHODOLOGY

Requirement Analysis:

In this stage, system requirements were identified by analyzing user needs and the recruitment process. The objective of this phase was to understand what functionalities the system should provide and how users would interact with the system. Information was gathered by studying existing job portals and identifying the features required for efficient job management and application handling.

Different types of users were identified, including **Job Seekers**, **Recruiters**, and **Administrators**, each having specific roles and permissions within the system. Functional requirements such as user registration, login, job posting, job search, application submission, and resume upload were defined. Non-functional requirements such as system security, performance, reliability, and usability were also considered.

This stage helped in clearly defining the system scope and establishing a foundation for designing and developing the system.

System Design:

In the system design phase, the overall structure of the Job Portal System was planned and documented. The system architecture was designed using a **3-tier architecture model**, consisting of the **Frontend**, **Backend**, and **Database** layers. This design ensures better system organization, scalability, and maintainability.

The user interface layout was designed to provide a simple and user-friendly experience for users. Navigation menus, forms, and web pages were planned to allow users to easily access system features. The database schema was also designed during this stage to define how data would be stored, retrieved, and managed.

Tables such as **Users**, **Jobs**, **Applications**, and **Admin** were created to store system data. Relationships between tables were defined to ensure data consistency and efficient data retrieval. The system workflow and data flow between components were also established in this phase.

This stage provided a blueprint for the development of the system.

Development:

In the development phase, the actual implementation of the Job Portal System was carried out based on the system design. The system was developed using web technologies for both frontend and backend components, and a relational database was used for data storage.

The frontend interface was developed using **HTML** and **CSS** to create responsive web pages that allow users to interact with the system. Pages such as login, registration, job listing, job posting, and application submission were created. The design focused on providing a clear and user-friendly interface.

The backend functionality was implemented using **Spring Boot**, which handled the system logic and processed user requests. Backend modules were developed to manage user authentication, job posting, job searching, and application processing. The backend also handled communication between the frontend and the database.

Database connectivity was established using **MySQL**, where tables were created to store user information, job details, and application records. Data validation and error handling mechanisms were implemented to ensure reliable system performance.

This phase resulted in a fully functional system with integrated frontend, backend, and database components.

Testing:

The testing phase was conducted to ensure that the Job Portal System functions correctly and meets all specified requirements. Different types of testing were performed to identify and fix errors before deployment.

Unit testing was performed to verify the functionality of individual modules such as login, registration, job posting, and application submission. Each module was tested separately to ensure correct behavior.

Integration testing was carried out to ensure smooth communication between system components, including frontend-to-backend and backend-to-database interactions. This testing verified that data was transmitted and processed correctly across the system.

System testing was conducted to evaluate the performance of the complete system under real-world conditions. The system was tested for user authentication, job searching, job posting, resume uploading, and application submission processes.

Error handling and validation testing were also performed to ensure the system responds appropriately to invalid inputs, missing data, and system failures. This phase helped in improving system reliability and ensuring that the system operates without errors.

Deployment:

In the deployment phase, the completed Job Portal System was installed and configured to make it available for users. The application was deployed on a **local server environment** for demonstration and evaluation purposes.

The server configuration included setting up the application environment, connecting the system to the MySQL database, and ensuring proper system functionality. Final testing was performed after deployment to verify that the system was accessible and working correctly.

The deployed system allowed users to register, log in, post jobs, search for jobs, and submit applications through the web interface. This phase marked the successful completion of the system development process and made the application ready for presentation and practical use.

CHAPTER 8 EXPERIMENTATION

The Job Portal System was tested using various scenarios to verify system performance and functionality.

Test Case 1 — User Registration

Input: User details

Output: Account created successfully

Test Case 2 — User Login

Input: Email and password

Output: Login successful

Test Case 3 — Post Job

Input: Job details

Output: Job added to database

Test Case 4 — Apply Job

Input: Resume file

Output: Application submitted

Test Case 5 — Download Resume

Input: Application ID

Output: Resume downloaded

These test cases confirmed that the system performs correctly under different conditions.

CHAPTER 9 RESULT AND ANALYSIS

The Job Portal System was successfully developed and tested. All modules functioned correctly without errors. Users were able to register, log in, post jobs, apply for jobs, upload resumes, and manage applications.

The system demonstrated high performance and reliability. Data was stored securely in the database, and user requests were processed quickly. The user interface was simple and easy to use, improving user satisfaction.

Performance Metrics:

System Response Time: Fast

Data Accuracy: High

System Reliability: High

User Satisfaction: High

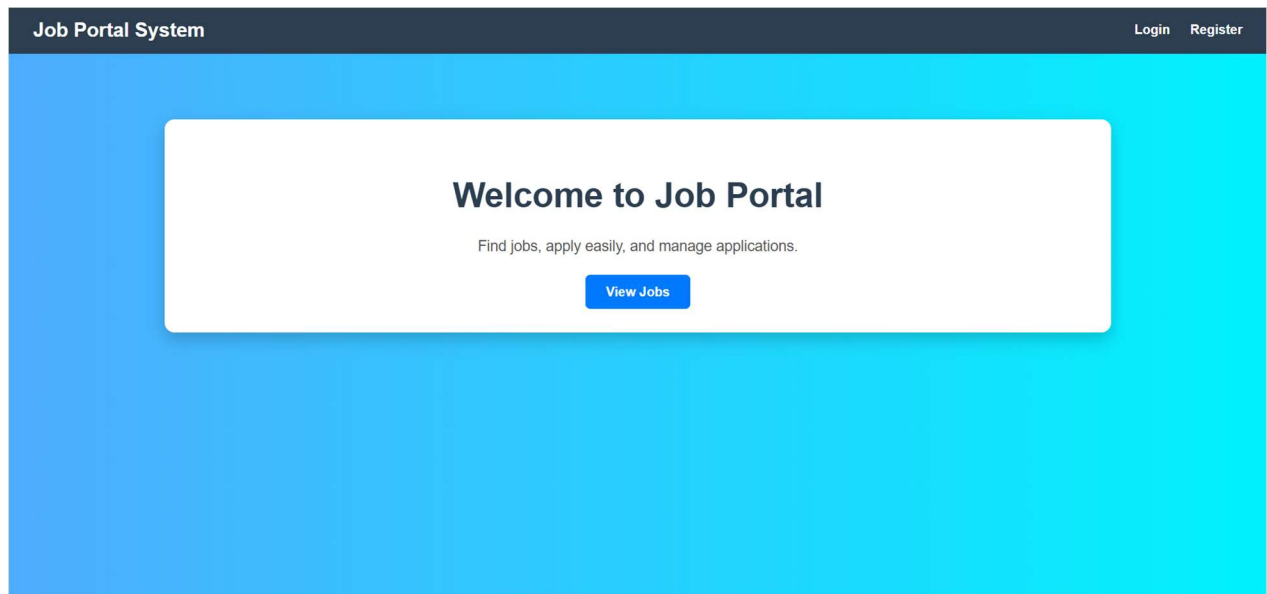


Fig 2:view of website

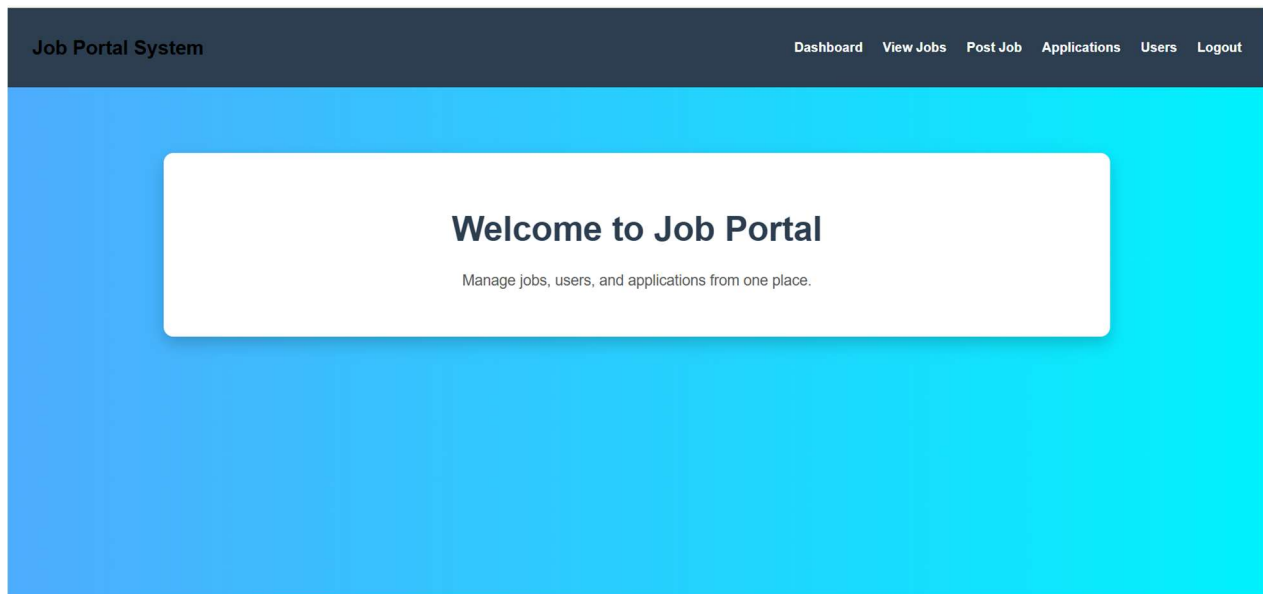


Fig 3: dashboard view

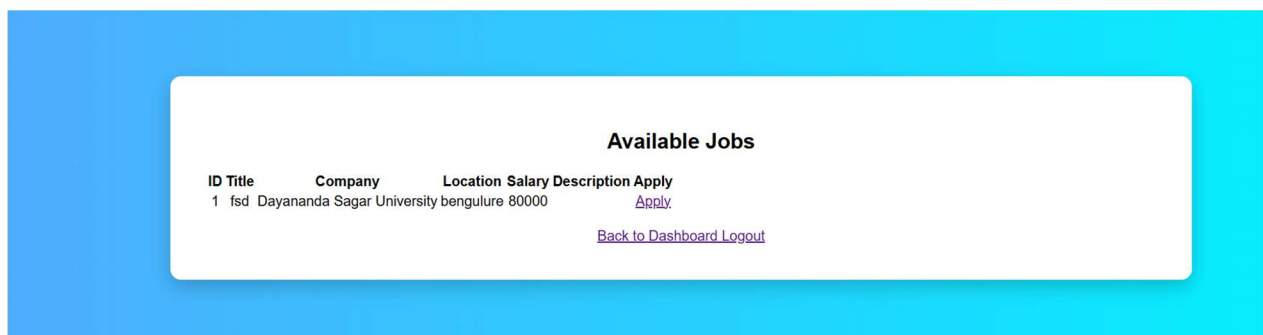


Fig 4 : jobs for applying

Registered Users					
		Search by name		Search	
ID	Name	Email	Password	Role	Action
1					Edit Delete
2					Edit Delete
3	puneeth	puneethreddyporeddy2006@gmail.com	12345	student	Edit Delete
6	aakash	akashsadhu967@gmail.com	76543	student	Edit Delete
7	aakash	akashsadhu967@gmail.com	87654	student	Edit Delete
8	aakash	akashsadhu967@gmail.com	5432	student	Edit Delete
9	puneeth	puneethreddyporeddy2006@gmail.com	6543	student	Edit Delete
10	puneeth	puneethreddyporeddy2006@gmail.com	12345	student	Edit Delete
11	ravi	eng24am0075@dsu.edu.in	Ravi@2006	student	Edit Delete
12	aakash	akashsadhu967@gmail.com	1234	student	Edit Delete
13	puneeth	puneethreddyporeddy2006@gmail.com	123456	student	Edit Delete
14	rghj	akashsadhu967@gmail.com	23456	student	Edit Delete
Back to Dashboard Logout					

Fig 5:admin view

The results indicate that the system meets the requirements and provides an effective solution for managing recruitment processes.

CHAPTER 10 CONCLUSION AND FUTURE WORK

The Job Portal System successfully automates the recruitment process by providing a digital platform for job seekers and employers. The system simplifies job searching, application submission, and resume management.

The project demonstrates the integration of frontend, backend, and database technologies in a full-stack web application. It improves efficiency, reduces manual work, and ensures secure data storage. Overall, the system provides a practical and scalable solution for managing recruitment processes in modern organizations.

Future work :

The Job Portal System can be enhanced with additional features to improve functionality and user experience. Possible future improvements include:

Email notifications for job applications

- Role-based login system

- Job search filters based on location and salary

- Mobile application development

- Cloud deployment

- Online interview scheduling

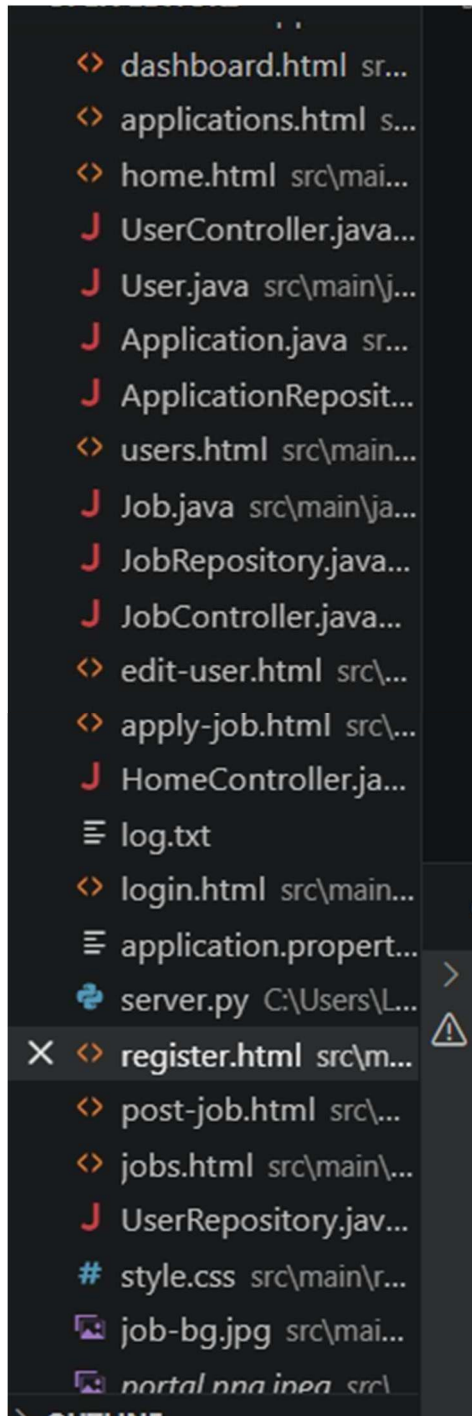
- Artificial Intelligence-based job recommendations

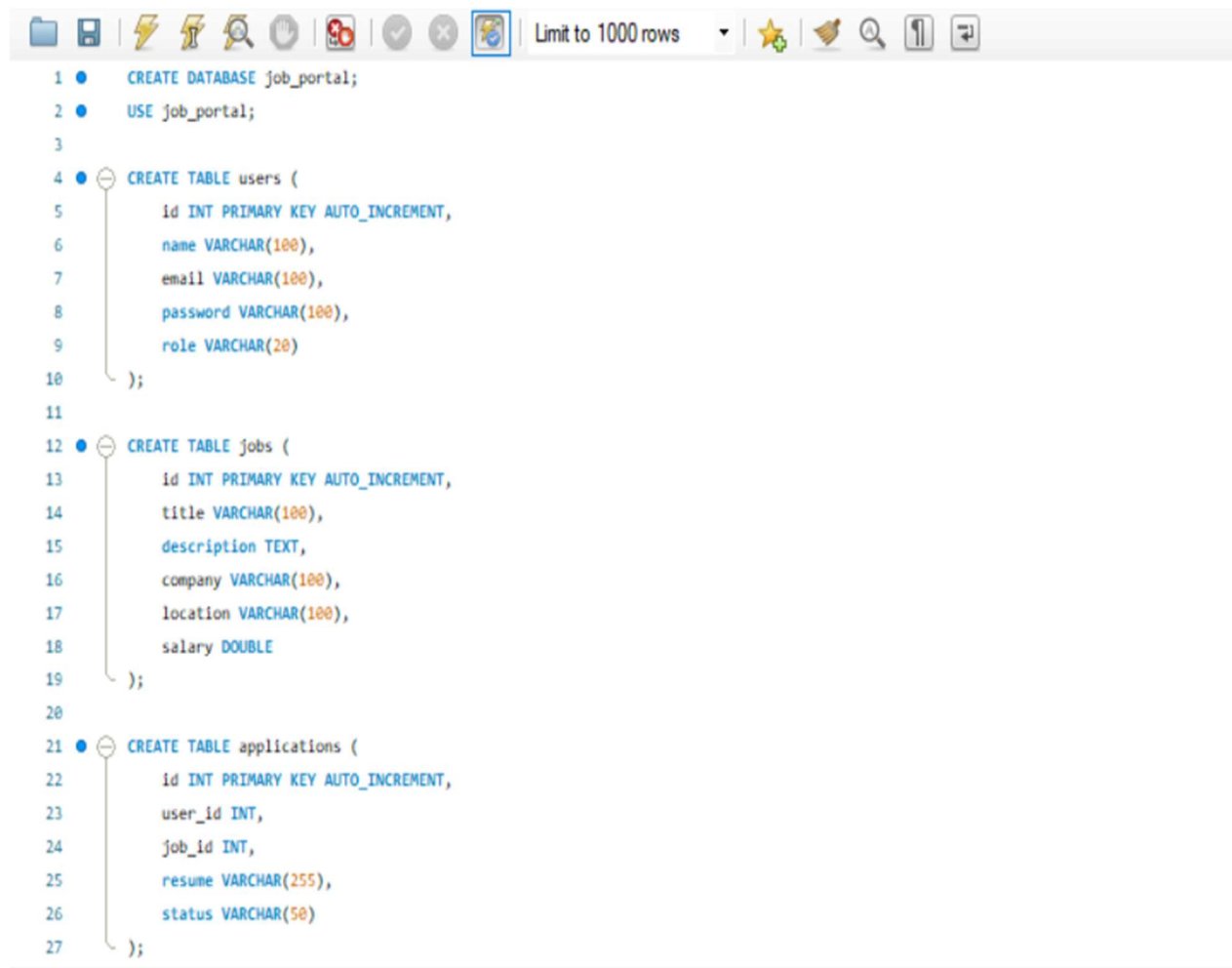
These enhancements will make the system more powerful and suitable for large-scale deployment.

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APPENDIX :





The image shows a screenshot of a database management tool's SQL editor. The interface includes a toolbar at the top with icons for file operations, execution, and search. Below the toolbar, the SQL code is displayed with line numbers from 1 to 27. The code defines a database named 'job_portal' and creates three tables: 'users', 'jobs', and 'applications'. The 'users' table has columns for id, name, email, password, and role. The 'jobs' table has columns for id, title, description, company, location, and salary. The 'applications' table has columns for id, user_id, job_id, resume, and status. The code is formatted with syntax highlighting, and the editor has a 'Limit to 1000 rows' option visible in the toolbar.

```
1  CREATE DATABASE job_portal;
2  USE job_portal;
3
4  CREATE TABLE users (
5      id INT PRIMARY KEY AUTO_INCREMENT,
6      name VARCHAR(100),
7      email VARCHAR(100),
8      password VARCHAR(100),
9      role VARCHAR(20)
10 );
11
12 CREATE TABLE jobs (
13     id INT PRIMARY KEY AUTO_INCREMENT,
14     title VARCHAR(100),
15     description TEXT,
16     company VARCHAR(100),
17     location VARCHAR(100),
18     salary DOUBLE
19 );
20
21 CREATE TABLE applications (
22     id INT PRIMARY KEY AUTO_INCREMENT,
23     user_id INT,
24     job_id INT,
25     resume VARCHAR(255),
26     status VARCHAR(50)
27 );
```

Fig 6 : database code

Result Grid					
Filter Rows:		Edit:			
Export/Import:		Wrap Cell Content:			
id	email	job_id	name	resume	
4	akashsadhu967@gmail.com	987654	ertee	NULL	
5	akashsadhu967@gmail.com	245	aakash	project (1).pptx	
6	akashsadhu967@gmail.com	23456	aakash	final fsd reportt.pdf	
7	akashsadhu967@gmail.com	125678	ravi	puneeth fsd.pdf	
8	akashsadhu967@gmail.com	12345	aakash	raviposter.pdf	
NULL	NULL	NULL	NULL	NULL	

Fig 7 :backend storage